

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0984

Moreno Tap 230 kV
January 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0984. The Interconnection Customer proposes a 105.4 MW solar photovoltaic facility interconnecting at the Moreno Tap 230 kV substation in the Ohio area. The request was received on April 22, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 323 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,074,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Myrtle Point - Mcmillan 230 KV	Myrtle Pt Substation - Post Rock 230 KV Line	112.6	524	12.3
Eastown Road Substation 100/1 kV Autotransformer	Swan Vly - Trego 500 KV L	106.3	213	5.9
Portage Lakes - Weedpatch 230 KV	Trego - Beaver 500 KV	104.2	875	5.5
Swan Valley - Trego 500 KV line	Troy Energy Llc - Trego 500 KV	104.0	4,453	18.9
Burris Pit - James De Young 230 KV Line	Trego - Beaver 500 KV	102.5	1,392	7.6
Baaga Vlg 500-230 kV Autotransformer	Beaver - Trego 500 KV	101.5	1,689	11.7
Madeline - E Methuen 230 KV	Gray'S Flat - E Methuen 230 KV	101.5	1,597	8.3
Hebo - Barrett 230 KV	Winthrop - Hebo 230 KV	100.3	930	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcmillan - Myrtle Point 230 KV	Rebuild line with higher-capacity conductor	\$8,252,000	100% (\$8,252,000)
Eastown Road Substation 100/1 kV Autotransformer	Replace transformer with higher-rated unit	\$3,472,000	100% (\$3,472,000)
Portage Lakes - Weedpatch 230 KV	Reconductor line	\$12,243,000	21% (\$2,607,000)
Swan Vly - Trego 500 KV	Reconductor line	\$52,422,000	1% (\$754,000)
Burris Pit - James De Young 230 KV Line	Reconductor line	\$14,736,000	2% (\$249,000)
Baraga Vlg 500-230 kV XFMR	Replace transformer with higher-rated unit	\$34,406,000	1% (\$493,000)
Madeline - E Methuen 230 KV Line	Reconductor line	\$15,489,000	1% (\$198,000)
Hebo - Barrett 230 KV	Reconductor line	\$2,819,000	2% (\$49,000)
	Total allocated to project		\$16,074,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.